

FICTIONAL DEMO SUBMISSION

For LMS grading-workflow demonstration only

Assignment 1 - Function Transformations

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Selected responses

These original responses are intentionally concise and are provided only to demonstrate Moodle submission, download, and grading workflows.

Question 1

The negative leading factor reflects the parabola and makes it narrower. I used the point mapping $(x, y) \rightarrow (x + 3, -2y + 5)$. The parent points $(-1,1)$, $(0,0)$, and $(1,1)$ become $(2,3)$, $(3,5)$, and $(4,3)$.

Question 3

For $r(x) = -4/[2(x - 1)] + 3$, I simplify the scale to $-2/(x - 1) + 3$. Relative to $1/x$: $a = -4$, $k = 2$, $d = 1$, $c = 3$. The vertical asymptote is $x = 1$ and the horizontal asymptote is $y = 3$.

Question 5

I used $y = a(x - 2)^2 + 6$ and substituted $(4, -2)$: $-2 = 4a + 6$, so $a = -2$. The function $y = -2(x - 2)^2 + 6$ opens downward, has maximum $(2,6)$, and passes through $(4,-2)$.